SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

ASSESSMENT TOOL 3 – Tree Traversal-Based Questions

Course Code:	CSA03	**Course Title:**	Data Structures
CO Assessed:	CO3 – Imitation (BL3, BL4)	**Assessment:**	Assessment Tool 3- Tree Traversal Based Questions
Weightage:	10% of CIA	**Total Marks:**	50 (5 Questions × 10 Mark)
CO5 Statement:	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity (BL3, BL4)		
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Section / Batch:	2024-A	**Date:**	8-8-26

Code of Conduct

I K.Deeksha Sree(192424097)___(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K.Deeksha Sree_____________

Instructions

- This test contains 5 Traversal Questions Total: 50 Marks.
- Evaluation of Answer Quality - Understanding of Tree Structure, Traversal Execution, Procedure, Accuracy & Logical Reasoning and Complexity Analysis
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Annexure A – Traversal Based Questions

Q1. Insert the following elements into an empty Binary Search Tree:
45, 25, 15, 35, 65, 55, 75. Draw the resulting BST and find its:Preorder, Inorder, Postorder
Level-order traversal
Q2. A binary tree has the following traversals:
Inorder: D B E A F C G
Preorder: A B D E C F G
Construct the binary tree and determine its postorder traversal.
Q3. A Binary Search Tree contains the keys:
40, 20, 60, 10, 30, 50, 70, 25, 35
Analyze the four traversal techniques
Q4. Construct a binary tree from:
Preorder: 1 2 4 5 3 6 7
Inorder: 4 2 5 1 6 3 7
find the postorder and level-order traversals.
Q5. Construct a binary tree from:
Inorder: H D I B E A F C G
Postorder: H I D E B F G C A
Find the preorder traversal.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Tree Structure	20	Correctly understands and represents the given tree	Minor representation/interpretation error	Partially understands the structure	Unable to represent the tree correctly
Traversal Execution	20	Correctly performs all required traversals with proper node sequence	One minor error in traversal sequence	Multiple errors but approach is evident	Traversal procedure is incorrect
Algorithm / Procedure	20	Clearly explains traversal steps/algorithm	Explanation is mostly correct	Basic steps are given but incomplete	No clear algorithm/procedure
Clearly explains traversal steps/algorithm	20	Explanation is mostly correct	Basic steps are given but incomplete	No clear algorithm/procedure	Clearly explains traversal steps/

					algori thm
Complexity Analysis	20	Correctly identifies time and space complexity and justifies it	Correct complexity with limited explanation	Partially correct complexity	Incorrect or missing complexity

Faculty In-charge	Course Coordinator	Program Director
Signature: ___________	Signature: ___________	Signature: ___________
Date: _______________	Date: _______________	Date: _______________

1.
Elements:
45, 25, 15, 35, 65, 55, 75
Step 1: Insert 45
45 becomes the root.

```
    45
```

Step 2: Insert 25
25 < 45 → go to the left.

```
      45
     /
   25
```

Step 3: Insert 15
15 < 45 → left
15 < 25 → left

```
      45
     /
   25
   /
 15
```

Step 4: Insert 35
35 < 45 → left
35 > 25 → right

```
      45
     /
   25
   / \
 15  35
```

Step 5: Insert 65
65 > 45 → right.

```
       45
      /  \
    25    65
   / \
 15  35
```

Step 6: Insert 55
55 > 45 → right
55 < 65 → left
Step 7: Insert 75
75 > 45 → right
75 > 65 → right
Final BST

```
          45
        /    \
      25      65
     /  \    /  \
   15   35  55   75
```

Traversals

1. Preorder = Root → Left → Right
45 → 25 → 15 → 35 → 65 → 55 → 75
Preorder: 45, 25, 15, 35, 65, 55, 75

2. Inorder = Left → Root → Right
15 → 25 → 35 → 45 → 55 → 65 → 75
Inorder: 15, 25, 35, 45, 55, 65, 75
Notice: In a BST, **inorder always gives sorted order**.

3. Postorder = Left → Right → Root
15 → 35 → 25 → 55 → 75 → 65 → 45
Postorder: 15, 35, 25, 55, 75, 65, 45

4. Level-order = Level by level
45 → 25 → 65 → 15 → 35 → 55 → 75
Level-order: 45, 25, 65, 15, 35, 55, 75
Final Answer — Q1

Traversal	**Result**
Preorder	**45, 25, 15, 35, 65, 55, 75**
Inorder	**15, 25, 35, 45, 55, 65, 75**
Postorder	**15, 35, 25, 55, 75, 65, 45**
Level-order	**45, 25, 65, 15, 35, 55, 75**

Q2. Construct Binary Tree
Given:
Inorder:
D B E A F C G
Preorder:
A B D E C F G
Step 1: Find root
In **preorder**, the first element is always the root.
So:
Root = A
Find A in inorder:
D B E | A | F C G
Left subtree:
D B E
Right subtree:
F C G
Step 2: Construct left subtree
Preorder remaining:
B D E C F G
The next element is **B**, so B is the left child of A.
Inorder:
D | B | E
Therefore:
D = left child of B
E = right child of B

```
    A
   /
  B
 / \
D   E
```

Step 3: Construct right subtree
Right inorder:
F C G
Next preorder element is **C**.

So C is the right child of A.
Inorder:
F | C | G
Therefore:
F = left child of C
G = right child of C

```
      A
     / \
    B   C
   / \ / \
  D  E F  G
```

Postorder
Postorder = **Left → Right → Root**
Left subtree:
D E B
Right subtree:
F G C
Then root:
A
Postorder = D, E, B, F, G, C, A
Q3. BST Traversals
Given keys:
40, 20, 60, 10, 30, 50, 70, 25, 35
Construct the BST.
Insertion
40 → root
20 < 40 → left
60 > 40 → right
10 < 40 → < 20 → left of 20
30 < 40 → > 20 → right of 20
50 > 40 → < 60 → left of 60
70 > 40 → > 60 → right of 60
25 < 40 → > 20 → < 30 → left of 30
35 < 40 → > 20 → > 30 → right of 30
BST

```
             40
           /    \
         20      60
        /  \    /  \
      10   30  50   70
          /  \
         25  35
```

Now let's analyze the **four traversal techniques**.
1. Preorder
Rule:
Root → Left → Right
40 → 20 → 10 → 30 → 25 → 35 → 60 → 50 → 70
Preorder: 40, 20, 10, 30, 25, 35, 60, 50, 70

2. Inorder
Rule:
Left → Root → Right
10 → 20 → 25 → 30 → 35 → 40 → 50 → 60 → 70
Inorder: 10, 20, 25, 30, 35, 40, 50, 60, 70
Again, because this is a BST, the result is **sorted**.
3. Postorder
Rule:
Left → Right → Root
10 → 25 → 35 → 30 → 20 → 50 → 70 → 60 → 40

Postorder: 10, 25, 35, 30, 20, 50, 70, 60, 40

4. Level-order

Visit each level from top to bottom:

Level 1: 40

Level 2: 20 60

Level 3: 10 30 50 70

Level 4: 25 35

Therefore:

Level-order: 40, 20, 60, 10, 30, 50, 70, 25, 35

Final Answer — Q3

Traversal	Result
Preorder	**40, 20, 10, 30, 25, 35, 60, 50, 70**
Inorder	**10, 20, 25, 30, 35, 40, 50, 60, 70**
Postorder	**10, 25, 35, 30, 20, 50, 70, 60, 40**
Level-order	**40, 20, 60, 10, 30, 50, 70, 25, 35**

Q4. Construct Binary Tree

Given:

Preorder:

1 2 4 5 3 6 7

Inorder:

4 2 5 1 6 3 7

Step 1: Root

First element of preorder = **1**

Inorder:

4 2 5 | 1 | 6 3 7

So:

Left subtree = 4 2 5

Right subtree = 6 3 7

Step 2: Left subtree

Next preorder element = **2**

Inorder left part:

4 | 2 | 5

Therefore:

```
  2
 / \
4   5
```

Step 3: Right subtree

Next root = **3**

Inorder:

6 | 3 | 7

Therefore:

```
  3
 / \
6   7
```

Final tree

```
      1
     /  \
    2    3
   / \  / \
  4  5 6   7
```

Postorder

Left → Right → Root

Left subtree:

4 → 5 → 2
Right subtree:
6 → 7 → 3
Root:
1
Postorder = 4, 5, 2, 6, 7, 3, 1
Level-order
Level by level:
1
2 3
4 5 6 7
Level-order = 1, 2, 3, 4, 5, 6, 7
Q5. Construct Binary Tree
Given:
Inorder:
H D I B E A F C G
Postorder:
H I D E B F G C A
We need to find **Preorder**.
Step 1: Find root
In postorder, the **last element is the root**.
So:
Root = A
Inorder:
H D I B E | A | F C G
So:
Left subtree = H D I B E
Right subtree = F C G
Step 2: Left subtree
Postorder for left subtree:
H I D E B
Last element = **B**
So B is the left child of A.
Inorder:
H D I | B | E
Therefore:
Left of B = H D I
Right of B = E
For H D I, postorder:
H I D
Root = **D**
Inorder:
H | D | I
So:
```
   D
  / \
 H   I
```
Thus left side becomes:
```
      B
     / \
    D   E
   / \
  H   I
```
Step 3: Right subtree
Right inorder:
F C G
Right postorder:
F G C
Last element = **C**

So C is the right child of A.
Inorder:
F | C | G
Therefore:

```
  C
 / \
F   G
```

Final tree

```
      A
     /  \
    B    C
   / \  / \
  D   E F   G
 / \
H   I
```

Find Preorder
Preorder = **Root → Left → Right**
Start at A:
A
Then left subtree:
B → D → H → I → E
Then right subtree:
C → F → G
Therefore:
Preorder = A, B, D, H, I, E, C, F, G